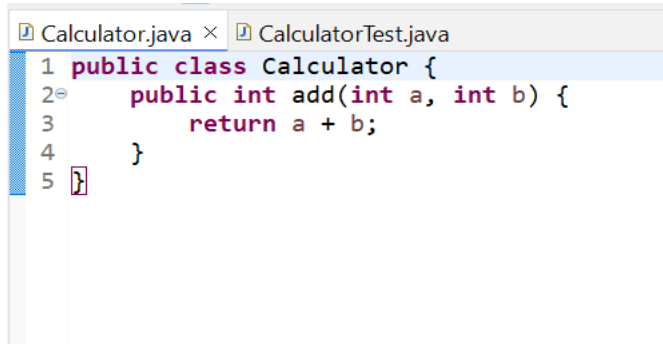


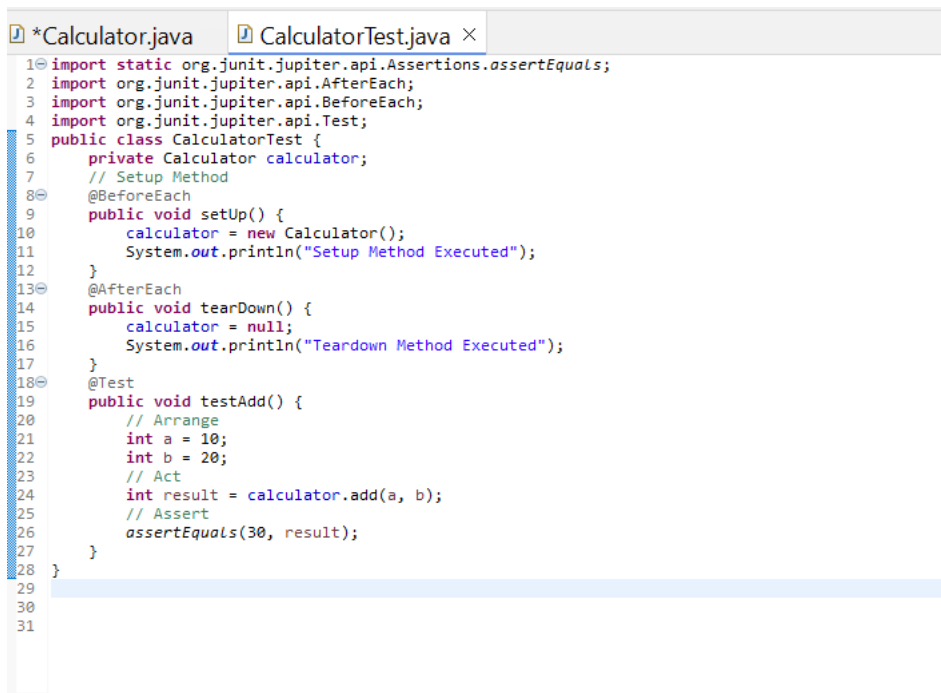
Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup, and Teardown Methods in Junit

Step 1: Create Calculator Class



```
1 public class Calculator {
2     public int add(int a, int b) {
3         return a + b;
4     }
5 }
```

Step 2: Create CalculatorTest Class



```
1 import static org.junit.jupiter.api.Assertions.assertEquals;
2 import org.junit.jupiter.api.AfterEach;
3 import org.junit.jupiter.api.BeforeEach;
4 import org.junit.jupiter.api.Test;
5 public class CalculatorTest {
6     private Calculator calculator;
7     // Setup Method
8     @BeforeEach
9     public void setUp() {
10         calculator = new Calculator();
11         System.out.println("Setup Method Executed");
12     }
13     @AfterEach
14     public void tearDown() {
15         calculator = null;
16         System.out.println("Teardown Method Executed");
17     }
18     @Test
19     public void testAdd() {
20         // Arrange
21         int a = 10;
22         int b = 20;
23         // Act
24         int result = calculator.add(a, b);
25         // Assert
26         assertEquals(30, result);
27     }
28 }
29
30
31
```

Step 3. Save Files

- Press **Ctrl + S**

Step 4. Run Test

- Right-click **CalculatorTest.java**
- **Run As → JUnit Test**

AAA Pattern

Arrange

Prepare test data.

int a = 10;

int b = 20;

Act

Call the method.

```
int result = calculator.add(a, b);
```

Assert

Verify the result.

```
assertEquals(30, result);
```

.....

Setup and Teardown

@BeforeEach

Runs before every test.

@BeforeEach

```
public void setUp() {  
    calculator = new Calculator();  
}
```

@AfterEach

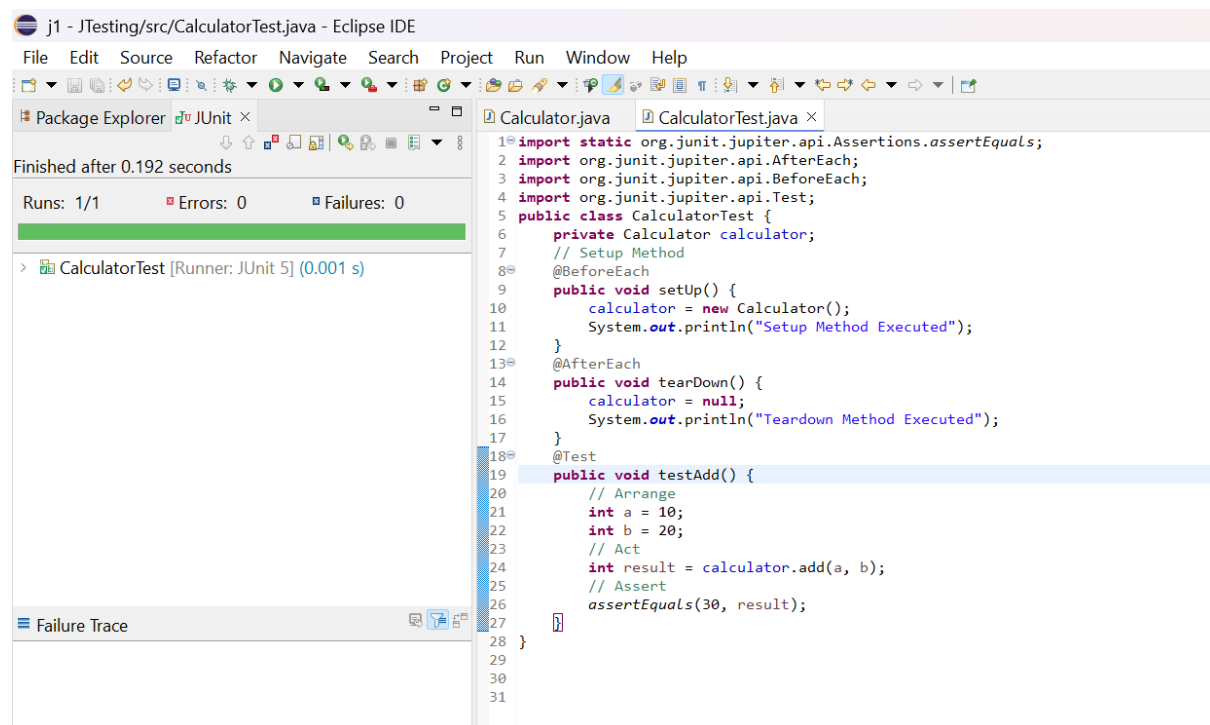
Runs after every test.

@AfterEach

```
public void tearDown() {  
    calculator = null;  
}
```

.....

Output



```
j1 - JTesting/src/CalculatorTest.java - Eclipse IDE  
File Edit Source Refactor Navigate Search Project Run Window Help  
Package Explorer JUnit x  
Finished after 0.192 seconds  
Runs: 1/1 Errors: 0 Failures: 0  
> CalculatorTest [Runner: JUnit 5] (0.001 s)  
Failure Trace  
1 import static org.junit.jupiter.api.Assertions.assertEquals;  
2 import org.junit.jupiter.api.AfterEach;  
3 import org.junit.jupiter.api.BeforeEach;  
4 import org.junit.jupiter.api.Test;  
5 public class CalculatorTest {  
6     private Calculator calculator;  
7     // Setup Method  
8     @BeforeEach  
9     public void setUp() {  
10         calculator = new Calculator();  
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24         int result = calculator.add(a, b);  
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26         assertEquals(30, result);  
27     }  
28 }  
29  
30  
31
```